# Perimeters and areas of similar figures

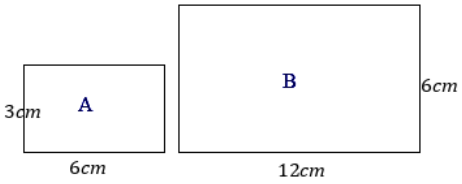
**1** Consider the two similar rectangles shown.

**a** Find the area of Rectangle A.

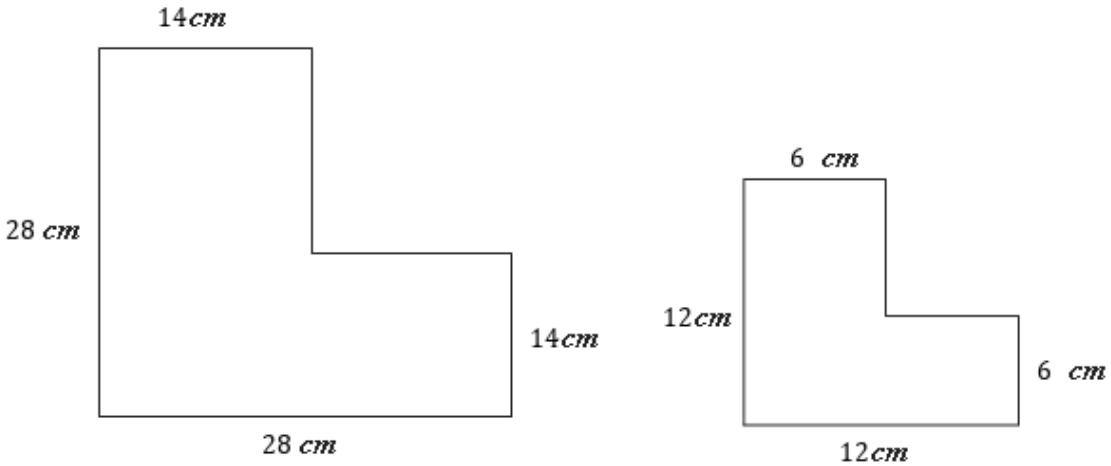
**b** Find the area of Rectangle B.

**c** What is the ratio of the area of Rectangle A to Rectangle B?

**d** Is it true in this case that if the matching sides of two similar figures are in the ratio $m : n$, then their areas are in the ratio $m^2 : n^2$?




**2** The corresponding sides of two similar figures are in the ratio $7 : 3$.

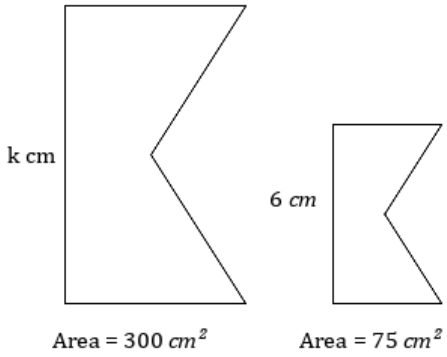



**a** State the length scale factor from the left to the right figure.

**b** State the area scale factor from the left to the right figure.

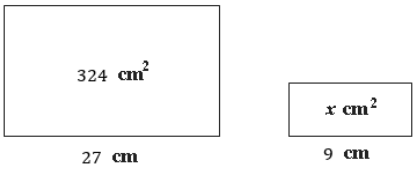
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**3** The figures in each pair are similar.
Find the value of $k$.

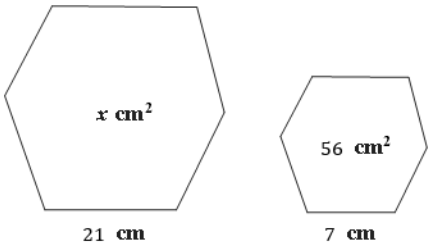



**4** Find the value of $x$ for each of the following figures:

**a**



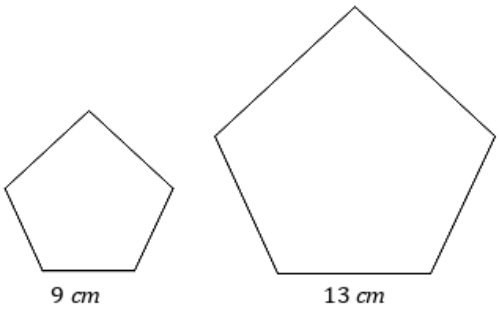

**b**




**5** Two similar pentagons have sides of length $9$ cm and $13$ cm.

**a** Find the ratio between the smaller area and the larger area.

**b** If the area of the smaller pentagon is $81$ $\text{cm}^2$, find the area of the larger pentagon.




**6** A triangle has side lengths of $7$ cm, $10$ cm and $16$ cm. A similar triangle to it has $9$ times the area of the first triangle.

**a** Find the length scale factor between the two triangles.

**b** Find the side lengths of the second triangle.

**c** State the relationship between the perimeters of the two triangles.

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**7** A square of side length $6\ \text{cm}$ is enlarged using the scale factor $3$. Find the area of the enlarged square.

**8** Two similar parallelograms have matching sides whose lengths are in the ratio $7 : 2$. Find the ratio of their areas.

**9** The diameter of a circle is tripled. What happens to its area?

**10** Consider two similar parallelograms with a matching sides in the ratio $6 : 8$.

- **a** Calculate the area of the larger parallelogram, if the area of the smaller parallelogram is $72\ \text{cm}^2$.
- **b** Find the length of the base of the smaller parallelogram , if the length of the base of larger parallelogram is $12\ \text{cm}$.

**11** The corresponding sides of two similar triangles are $8\ \text{cm}$ and $40\ \text{cm}$. If the area of the smaller triangle is $24\ \text{cm}^2$, find the area of the larger triangle.

**12** The cost of constructing a garden bed is dependent on the area it covers. A rectangular garden bed measuring $16\ \text{ft}$ by $24\ \text{ft}$ costs a known amount to construct. If you decided to create a similar garden bed, measuring $20\ \text{ft}$ by $30\ \text{ft}$, by what factor would the cost increase?

**13** Consider two similar ceilings where the first has dimensions $5\ \text{m}$ by $4\ \text{m}$, and the second has dimensions $20\ \text{m}$ by $16\ \text{m}$.

- **a** Find the length scale factor.
- **b** Find the area scale factor.
- **c** If the smaller ceiling took $1.5\ \text{L}$ of paint to cover it, determine the amount in liter of paint required to cover the second ceiling.

**14** Consider the formula $A = \frac{1}{2}bh$. Determine the effect of each of the following have on $A$.

- **a** The values of $b$ and $h$ double.
- **b** The value of $b$ increases by a factor of $2$ and the value of $h$ decreases by a factor of $2$.

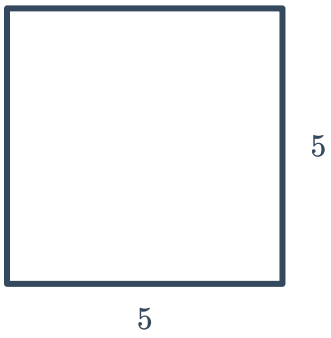
**15** Consider the formula $A = \pi r^2$. Determine the effect of each of the following have on $A$.

- **a** The value of $r$ triples.
- **b** The value of $r$ decreases by a factor of $4$.

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## Additional questions

**16** Emily is knitting a rectangular blanket. So far the blanket measures $5\ \text{ft} \times 5\ \text{ft}$. If Emily increases both the length and the width of the blanket by $1\ \text{ft}$, how many times bigger will the area of the finished blanket be?




**17** A triangle with an area of $\frac{3}{5}\ \text{m}^2$ is dilated by a factor of $10$. Find the area of the new triangle.

**18** A regular pentagon has been enlarged with a scale factor of $3.25$. Describe the relationship between the perimeter of the two pentagons.

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